

YUZHEN CHEN

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EDUCATION

Harvard University

M.S. in Computational Science and Engineering

Relevant Coursework: Video Generation & Multi-modality World Model & VLA

Cambridge, MA

Expected May 2027

University of Michigan

B.S.E. in Computer Engineering; GPA: 3.9/4.0

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Vision, Computer Architecture, Robotic Kinematics and Dynamics, Robot Perception, SLAM, Linear Algebra, Multivariable Calculus, Differential Equations, Discrete Math, Probability and Statistics, Digital Circuit Design, Logic Design, Embedded System, GPU Parallel Computation, Robot 3D Reconstruction, Human Computer Interactive

Ann Arbor, MI

May 2025

HONORS & AWARDS

OMPP Summer Research Fellowship | Harvard University

2026

Best Paper Award, Runner-Up | CVPR 2026 WMAS Workshop

2026

Spotlight | CVPR 2026 3D-LLM/VLA Workshop

2026

Best Poster Award | IROS 2023

2023

James B. Angell Scholar | University of Michigan

2023-2024

Dean's List | University of Michigan

2022-2025

RESEARCH EXPERIENCE

GEM-4D: Geometry-Enhanced Video World Models for Robot Manipulation [6]

Cambridge, MA

Research Assistant, Advisors: [Dr. Paul Pu Liang](#) (MIT Media Lab) & [Dr. Mengyu Wang](#) (Harvard AI and Robotics Lab)

Sept. 2025 - Present

- Developed a geometry-grounded video world model that distills dense 4D correspondence supervision into a video diffusion transformer during training, improving physical consistency with no added inference cost.
- Proposed an adaptive inverse dynamics system that converts generated videos into executable 6-DoF robot trajectories using vision foundation models and confidence-gated pose recovery.
- Achieved state-of-the-art 4D scene generation—improving RGB video quality (SSIM 76.2→82.1, FVD 33.2→31.8) and point-cloud reconstruction (Chamfer distance 0.244→0.200) over the best prior baselines—and raised real-world robot manipulation success from 61% to 81%.

Interpretable Glaucoma Diagnosis with Retinal Foundation Models and Medical VLMs

Boston, MA

Research Assistant, Advisor: [Dr. Mengyu Wang](#) (Harvard AI and Robotics Lab)

Jan. 2026 - Present

- Proposed an interpretable glaucoma diagnosis framework that maps retinal images to both a three-class diagnosis (*likely, borderline, not likely*) and a structured six-step clinical reasoning trace for more explainable and clinically traceable predictions.
- Designed the CoT supervision strategy and led the curation of 700+ expert-annotated retinal images, creating a specialized dataset of step-wise glaucoma diagnostic rationales for supervised VLM fine-tuning.
- Developed a RETFound-MedGemma fine-tuning pipeline with class-weighted LoRA adaptation, improving three-way diagnosis accuracy from 49.4% to 70.2% while generating interpretable diagnostic rationales.

Act2See: VLM-Guided Active Exploration for Online URDF Reconstruction

Boston, MA

Research Assistant, Advisor: [Dr. Mengyu Wang](#) (Harvard AI and Robotics Lab)

Jan. 2026 - Present

- Developed a VLM-guided active exploration framework that enables a robot to autonomously inspect an unknown articulated object, select promising parts or joint hypotheses to test, choose whether to manipulate, confirm, or skip each target, and stop once the object structure is sufficiently resolved.
- Designed a streaming online reconstruction pipeline that updates an explicit URDF frame-by-frame as RGB-D observations arrive during interaction, without waiting for the full manipulation sequence to finish; the maintained URDF incrementally stores part geometry, joint axes, joint types, and articulation states.
- Proposed a reconstruct-while-moving module that estimates revolute and prismatic joint parameters from sparse visual tracks, converts motion cues into dense part masks, and performs pose-aligned fusion to refine the URDF while the object is being manipulated.

Multi-Modal Semantic Perception in Robotic Navigation [4], [5]

Ann Arbor, MI

Research Assistant, Advisor: [Dr. Ram Vasudevan](#)

Apr. 2023 - May 2025

- Modified Kinect Fusion to integrate material-based color mapping into 3D reconstruction, enabling real-time visualization of surface friction coefficients.
- Optimized material classification model using MiT-B1 architecture, achieving 4x faster inference (9 FPS) while maintaining accuracy across 6 material classes.
- Developed a voting pipeline combining SAM and SegFormer models, significantly reducing high-noise issues at object junctions in real-time material predictions.

Learning Physical Object Properties from Images Using Diffusion Model [2]

Ann Arbor, MI

Research Assistant, Advisor: [Dr. Arpan Kusari](#)

Sept. 2024 - May 2025

- Developed an automated pipeline using Meta MatSynth and Replica datasets to generate large-scale training data with varied material properties, lighting conditions, and camera angles.
- Trained an encoder to extract material properties (base color, metallic, roughness, reflectance, anisotropy, transmission) from images, enabling accurate Physically-Based Rendering function generation.
- Implemented Monte-Carlo differentiable rendering as decoder, achieving robust image reconstruction.

Conformalized Reachable Sets for Obstacle Avoidance With Spheres (CROWS)

Ann Arbor, MI

Research Assistant, Advisor: [Dr. Ram Vasudevan](#)

Apr. 2024 - Sept. 2024

- Developed ML models for CROWS, a safe motion planning system using conformalized neural networks for sphere parameter prediction, reducing planning time from 0.51s to 0.21s compared to ARMTD.
- Generated extensive training datasets through simulation-based sampling of motion trajectories, enabling robust training of three distinct neural networks for center, radii, and gradient predictions.
- Validated system effectiveness through successful collision-free motion demonstrations in both simulation and hardware experiments, ensuring safe robot operation in unstructured environments.

Real-Time Modeling for Elastic Rods [1], [3]

Ann Arbor, MI

Research Assistant, Advisor: [Dr. Ram Vasudevan](#)

Nov. 2023 - Sept. 2024

- Used DEFORM, a physics-based modeling system for real-time simulation of elastic rods, achieving accurate prediction of rope configurations in 3D space based on physical parameters.
- Implemented a motion capture pipeline for collecting ground truth data, facilitating model training, validation, and performance optimization through real-world rope behavior analysis.
- Applied the simulation framework to industrial robotics, enabling automated wire manipulation tasks including insertion, removal, and assembly operations.

SLIM-VDB: Real-Time 3D Probabilistic Semantic Mapping [7]

Ann Arbor, MI

Research Assistant, Advisor: [Dr. Katie Skinner](#)

May 2024 - Sept. 2024

- Contributed to a real-time 3D probabilistic semantic mapping system that leverages OpenVDB for memory-efficient volumetric scene representation.
- Integrated closed-set and open-set semantic predictions with RGB-D / point-cloud observations through probabilistic fusion to generate interpretable 3D semantic maps.
- Built visualization and evaluation tools for benchmark mapping experiments, supporting analysis of semantic quality, integration time, and memory efficiency.

INTERNSHIPS

Bardor | Software & System Engineer

May - Sept. 2025

- Bridging software/hardware requirements between Detroit clients and Chinese manufacturers, securing projects for instrument clusters and infotainment displays across Chrysler, Jeep, and Polaris platforms.

UMich Shapiro Design Lab | Educational Technology Developer

May 2024 - May 2025

- Designed and built a multi-screen dice with six independent displays that respond to motion, enabling dynamic group learning in language education and serving as a platform for game design exploration - deployed at university language centers and gaming design courses, presented at Serious Play Conference 2025.

PROJECTS

Embodied Intelligence Robot – ROVER | Independent Project

May. 2025 - Present

- Engineered a high-performance 5-DOF robotic arm achieving 1-meter reach with 1kg payload capacity, delivering 3x better cost-effectiveness (<\$2000) than commercial alternatives
- Achieved 50ms control loop latency through optimized ROS2 architecture and custom motion planning algorithms, enabling real-time multi-modal interaction with RGB-D SLAM, LiDAR, and natural language processing
- Served as a testing platform for research teams, accelerating VLA/VLM model deployment cycles through providing affordable, reproducible physical validation beyond simulation

PUBLICATIONS

1. Zhou, K.* , **Chen, Y.*** , Zhan, F., Hua, H., Chen, G., Chang, X., Qu, A., Du, Y., Liu, Z., Liang, P.P., Wang, M. (2026). [GEM-4D: Geometry-Enhanced Video World Models for Robot Manipulation](#). *Under review, ECCV 2026*. (*Equal contribution; Best Paper Runner-Up & Spotlight, CVPR 2026 Workshops)

2. Chen, Y., Wu, X., Zong, Y., **Chen, Y.**, Li, A., Wu, J., Zhang, B., Vasudevan, R. (2025). [DEFT: Differentiable Branched Discrete Elastic Rods for Modeling Furcated DLOs in Real-Time](#). *arXiv, 2025*.

3. **Chen, Y.**, Son, H., Kusari, A. (2025). [MatPredict: A Dataset and Benchmark for Learning Material Properties of Diverse Indoor Objects](#). *arXiv preprint* (under review).

4. Chen, Y., Zhang, Y., Brei, Z., Zhang, T., **Chen, Y.**, Wu, J., Vasudevan, R. (2024). [Differentiable Discrete Elastic Rods for Real-Time Modeling of Deformable Linear Objects](#). *CoRL 2024*, Munich.

5. Ewen, P., Gunjal, G., Chen, H., Li, A., **Chen, Y.**, Vasudevan, R. (2024). [Multi-Modal Semantic Perception Using Bayesian Inference](#). *IROS 2023*, Detroit. (**Best Poster Award**)

6. Ewen, P., Chen, H., **Chen, Y.**, Li, A., Bagali, A., Gunjal, G., Vasudevan, R. (2024). [You’ve Got to Feel It To Believe It: Multi-Modal Bayesian Inference for Semantic and Property Prediction](#). *RSS 2024*.

7. Sheppard, A., Ewen, P., Wilson, J., Sethuraman, A.V., Adewole, B., Li, A., **Chen, Y.**, Vasudevan, R., Skinner, K.A. (2026). [SLIM-VDB: A Real-Time 3D Probabilistic Semantic Mapping Framework](#). *IEEE RA-L*.

8. Oral Presentation: **Chen, Y.**, Cai, B. (2023) Development of high-performance but low-cost sensing node for intelligent monitoring system. *U-M UROP Symposium*, Ann Arbor, MI. (**Best Poster Award**)

LEADERSHIP POSITIONS

Robotics Community Event Organizer MIT CHIEF	2026 - Present
Teaching Assistant EECS442 Computer Vision; Physics 140X	Sept. 2022 - Apr. 2025
Departmental Ambassador ENGR 110; UMich Robotics Department	Sept. 2023 - Dec. 2024
Student Tutor UMich Engineering Center for Academic Success	Jan. - Apr. 2023

SKILLS

Programming Languages: Python, C++, C, CUDA, MATLAB, JavaScript, Shell Scripting, LaTeX

ML/DL Frameworks: PyTorch, TensorFlow, JAX, TensorRT, ONNX, Hugging Face, OpenCV, scikit-learn, NCNN

Robotics & Vision: ROS/ROS2, Gazebo, MoveIt, PCL, Open3D, SLAM (ORB-SLAM, LIO-SAM), Motion Planning (OMPL, ARMTD)

ML Models & Algorithms: Transformers (BERT, GPT, Whisper), Vision Models (YOLO, SAM, DINO), Diffusion Models, Neural Rendering (NeRF, 3D Gaussian Splatting), Reinforcement Learning (PPO, SAC)

Tools & Platforms: Docker, Git, Conda, WandB, TensorBoard, CUDA Toolkit, AWS/GCP, Weights & Biases

3D/CAD Software: Fusion360, Blender, SolidWorks, MeshLab, CloudCompare, Mimics

Languages: Mandarin Chinese (Native), English (Fluent)